

# Program overview

12-Sep-2023 12:44

**Year** 2023/2024  
**Organization** Mechanical, Maritime and Materials Engineering  
**Education** Minors Mechanical Engineering

| Code                                                 | Omschrijving                                                                                        | ECTS | p1 | p2 | p3 | p4 | p5 |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------|------|----|----|----|----|----|
| <b>Minors WB 2023</b>                                |                                                                                                     |      |    |    |    |    |    |
| <b>WB-MI-168-23</b>                                  | <b>WB-MI-168-23 Robotica</b>                                                                        |      |    |    |    |    |    |
| <b>Voor ET Bachelors</b>                             |                                                                                                     |      |    |    |    |    |    |
| IO3881-21                                            | Design in Robotics                                                                                  | 4    |    |    |    |    |    |
| TI3115TU                                             | Software Engineering Methods                                                                        | 5    |    |    |    |    |    |
| WB3700                                               | Robotics Minor Project                                                                              | 15   |    |    |    |    |    |
| WB3703                                               | Introduction to ROS                                                                                 | 2    |    |    |    |    |    |
| WB3705                                               | Dynamics for Minor Robotics                                                                         | 4    |    |    |    |    |    |
| <b>Voor IO Bachelors</b>                             |                                                                                                     |      |    |    |    |    |    |
| IO3881-21                                            | Design in Robotics                                                                                  | 4    |    |    |    |    |    |
| TI3105TU                                             | Intermediate Python Programming                                                                     | 5    |    |    |    |    |    |
| WB2632                                               | Advanced Engineering Design Project                                                                 | 6    |    |    |    |    |    |
| WB3700                                               | Robotics Minor Project                                                                              | 15   |    |    |    |    |    |
| <b>Voor WB Bachelors</b>                             |                                                                                                     |      |    |    |    |    |    |
| ET3033TU                                             | Circuit Analysis                                                                                    | 3    |    |    |    |    |    |
| IO3881-21                                            | Design in Robotics                                                                                  | 4    |    |    |    |    |    |
| TI3115TU                                             | Software Engineering Methods                                                                        | 5    |    |    |    |    |    |
| WB3168                                               | Robotic System Identification and Parameter Estimation                                              | 1    |    |    |    |    |    |
| WB3700                                               | Robotics Minor Project                                                                              | 15   |    |    |    |    |    |
| WB3703                                               | Introduction to ROS                                                                                 | 2    |    |    |    |    |    |
| <b>Voor TI Bachelors</b>                             |                                                                                                     |      |    |    |    |    |    |
| ET3033TU                                             | Circuit Analysis                                                                                    | 3    |    |    |    |    |    |
| IO3881-21                                            | Design in Robotics                                                                                  | 4    |    |    |    |    |    |
| WB3700                                               | Robotics Minor Project                                                                              | 15   |    |    |    |    |    |
| WB3703                                               | Introduction to ROS                                                                                 | 2    |    |    |    |    |    |
| WB3704                                               | Gazebo Assignment                                                                                   | 2    |    |    |    |    |    |
| WB3705                                               | Dynamics for Minor Robotics                                                                         | 4    |    |    |    |    |    |
| <b>WB-MI-217-23</b>                                  | <b>WB-MI-217-23 Engineering for Large-Scale Energy Conversion and Storage</b>                       |      |    |    |    |    |    |
| AE3516A                                              | Fundamentals of Wind Energy I                                                                       | 3    |    |    |    |    |    |
| ET3034TU                                             | Solar Energy                                                                                        | 3    |    |    |    |    |    |
| WB3575                                               | Energy Conversion: Devices, Systems and Efficiencies                                                | 4    |    |    |    |    |    |
| WB3580                                               | Energy Storage                                                                                      | 5    |    |    |    |    |    |
| WB3585                                               | Assessment of Energy Systems                                                                        | 3    |    |    |    |    |    |
| WB3595                                               | Design Project Renewables Based Energy Conversion and Storage                                       | 12   |    |    |    |    |    |
| <b>WB-MI-238-23 Minor Authenticity and Art Crime</b> |                                                                                                     |      |    |    |    |    |    |
| MS3310UL                                             | What is Authenticity in Art                                                                         | 5    |    |    |    |    |    |
| MS3320                                               | The Biography of Art Objects: Understanding the Material Life Cycle of Artwork through Physical and | 5    |    |    |    |    |    |
| MS3330EUR                                            | Crime and Disruption in the Art Market                                                              | 5    |    |    |    |    |    |
| MS3340UL                                             | Authenticity and Art Crime: A Challenging Case Study (Leiden part)                                  | 10   |    |    |    |    |    |
| MS3350                                               | Authenticity and Art Crime: A Challenging Case Study (Delft part)                                   | 5    |    |    |    |    |    |

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| <b>Year</b>         | <b>2023/2024</b>                                      |
| <b>Organization</b> | <b>Mechanical, Maritime and Materials Engineering</b> |
| <b>Education</b>    | <b>Minors Mechanical Engineering</b>                  |

**Minors WB 2023**

|                     |                                                       |
|---------------------|-------------------------------------------------------|
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| <b>WB-MI-168-23</b>                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Contact for students</b>               | Martin Klomp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Program Title</b>                      | Robotica                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>In association with the Faculty of</b> | 3mE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Program Goals</b>                      | <p>The student is able to develop a functional robot in a multidisciplinary team. This comprises the following technical competences:</p> <ul style="list-style-type: none"> <li>- executing a market analysis, designing the user experience (including external appearance), producing a slick robot (IO students already possess these competences)</li> <li>- calculating forces, torques, stiffness, strength, acceleration/deceleration, collisions and stability of the robot, and designing the mechanical parts and the drive train (WB students already possess these competences)</li> <li>- calculating currents, circuit dynamics, and heat production, selecting, reading, and interpreting sensor outputs, and developing, building and testing electronic circuits for power supply, safety, and data processing (EL students already possess these competences)</li> <li>- designing, implementing, and testing c++/python software modules for human interfacing, interaction, safety, and autonomous behavior of the robot (TI students already possess these competences).</li> </ul> <p>In addition, the students will learn how to implement the knowledge from their own BSc education in a large robotics project, and they learn to negotiate and discuss interdisciplinary trade-offs.</p> |
| <b>Intended for</b>                       | <p>Students from the following five TUD bachelor educations:</p> <ul style="list-style-type: none"> <li>- Mechanical Engineering</li> <li>- Industrial Design</li> <li>- Computer Science (Technische Informatica)</li> <li>- Electrical Engineering</li> <li>- Aerospace Engineering</li> </ul> <p>who want to build their own robot in a multidisciplinary team, and who want to learn some of the skills of their multidisciplinary teammates.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Gives access to</b>                    | This is not a schakelminor. After following this minor, you will be well-prepared for robotics-related MSc research in your own discipline. For example, after this minor, the Mechanical Engineering BSc student is recommended to take an MSc in Mechanical Engineering and work on a robotics-related thesis project.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Expected prior Knowledge</b>           | Using Linux and programming in python. This will be taught during the minor, but prior experience is useful.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Prerequisites Minor</b>                | At least 90 ECTS in one of the four BSc educations mentioned above. Additionally, strong motivation is required; admission is limited to 10 students per discipline, due to limited availability of expensive high-tech equipment and intensive coaching.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Minor Coherence / Goal</b>             | The students follow 15 ECTS of carefully selected multidisciplinary courses (no courses in your own discipline), to get acquainted with the specialization of your team members.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Minor Content 1</b>                    | <p>Each participant follows 15 ECTS of courses, and completes 15 ECTS of teamwork projects. The course list depends on your BSc background so you will not follow everything.</p> <p>WB3701: Design in Robotics (IDE)<br/> WB3702: Design in Robotics (ME/AE/CSE/EE)<br/> ET3033TU: Circuit Analysis (ME/AE/CSE/IDE)<br/> TI3115TU: Software Engineering Methods (ME/AE/EE)<br/> TI3105TU: Introduction to Python (IDE)<br/> WB3703: Introduction to ROS (all)<br/> WB3704: Gazebo Assignment (CSE)<br/> WB3700: Robotics Minor Project (all)<br/> WB3705: Dynamics for Minor Robotics (CSE/EE)<br/> WB2632: Project Mechanica (IDE)<br/> WB3168: Robot Mechanics (ME/AE)</p> <p>In addition, there are various tutorials on design tools, software packages, and production methods, all aligned with the main projects listed below (teamwork projects for all students):</p> <p>Robotics Minor Project</p> <p>During the projects you will also actively participate other activities:</p> <ul style="list-style-type: none"> <li>- (Online) meetup with coaches</li> <li>- (Online) meetup with customers</li> <li>- Participate (online) in the final demo day</li> </ul>                                                                                                                                       |
| <b>Education Methods</b>                  | Lectures, tutorials and projects. A team assignment is a huge part of this minor. So working in teams and flexibility is needed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Minor Assessment</b>                   | Each part is assessed separately. The minor is successfully completed if the requirements of the bachelor's programme of the student are met.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Maximum number of participants</b>     | 50                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Minimum number of participants</b>     | 20                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Minor Remarks / Schedule</b>           | <p>NOTE: Only 50 places are available. To be able to select students in order to create balanced teams, we have temporarily put the capacity at 70 students. After you enroll, we will ask you for a motivation letter by email. Thus, successful enrollment does not guarantee admission for this minor.</p> <p>Descriptions of the modules and the project can be found in the digital study guide.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

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**Voor ET Bachelors**

| IO3881-21                              | Design in Robotics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Course Coordinator</b>              | Dr.ing. M.C. Rozendaal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Contact Hours / Week</b><br>x/x/x/x |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Contents</b>                 | In this IDE module, which is embedded within the Minor Robotics, we have bundled core IDE competencies within a coherent teaching program. The module teaches students how human and business aspects relate to and integrate with technology development. An iterative design methodology, inspired by SCRUM and the Basic Design Cycle (BDC), will be followed to foster this integration. This Module provides students with a parallel track (that will continue throughout the semester) to help them learn to apply human and business aspects in their project, across the different stages of technology development (i.e., from the fuzzy front-end to optimization).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Study Goals</b>                     | <p>Students will learn how technology is used and experienced and by people and will learn how to position technology development in a business context. Further, students will be spurred on to adopt a critical perspective on technology. During the Minor these aspects are directly linked their prototyping activities that they will engage with throughout the Minor. Concerning methodology, students will learn about iterative design processes (based on SCRUM), and will be trained in how to communicate with others during this process and to collaborate in multidisciplinary setting that the Minor Robotics provides. Appropriate methods and tools from the Delft Design Guide (DDG) will be provided to students as a resource.</p> <p>We have developed 6 different tracks that will be explained in more detail below:</p> <ol style="list-style-type: none"> <li>1. Human track: Understanding how technology is used and experienced by people. Contextual Inquiry, socio-technical systems, user-system interaction, co-design, user testing.</li> <li>2. Business track: Understanding technology development in a business context. Value proposition, unique selling point, multistakeholder analysis, business modeling, entrepreneurship.</li> <li>3. Prototyping track: Understanding prototyping as a vehicle for human, business technology integration. Experience prototyping, low fi-modeling (foam), Wizard of Oz approaches, interaction design/formgiving.</li> <li>4. Critical thinking track: Understanding different viewpoints and values. Problem (re)framing, philosophy of technology, ethics.</li> <li>5. Collaboration &amp; Communication track: Understanding the importance of multidisciplinary teamwork. Teambuilding, presentation and communication, multidisciplinary collaboration, visualization.</li> <li>6. Design Process track: Understanding iterative design processes. SCRUM methodology, Basic Design Cycle</li> </ol> |   |
| <b>Education Method</b>                | This Module provides a parallel track in the Minor program. At different stages of their technology development, IDE knowledge and skills are provided that are appropriate and useful to them at that specific stage. The modules are organized in SCRUM sprints that are blocks of one or two weeks, in which a basic design cycle is followed. In each sprint a specific assignment is given to the students, that provides them with specific IDE knowledge and skills that they can directly apply.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Literature and Study Materials</b>  | Delft Design Guide, van Boeijen et al.<br>Integraal Product Ontwerpen, Roozenburg en Eekels                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Assessment</b>                      | The means in which the students learning is assessed differs between courses. Assessment is done by reviewing the design outcomes/deliverables (prototypes, individual reports, personal reflections) both formative and summative.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Special Information</b>             | Dr. ing. Marco Rozendaal, responsible teacher ID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |

| TI3115TU                               | Software Engineering Methods                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 5 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | T. Durieux                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/0/0 lecture; 4/0/0/0 lab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Exam Period</b>                     | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Expected prior knowledge</b>        | Basics of (Python) programming or currently taking TI3105TU.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Contents</b>                 | <p>In this course, students will learn:</p> <ul style="list-style-type: none"> <li>- software development processes, such as waterfall and agile</li> <li>- requirements engineering (use cases + user stories)</li> <li>- version control and Git</li> <li>- software testing</li> <li>- software design and design patterns</li> <li>- refactoring, code smells, and implementation best practices</li> </ul>                                                                                                                                          |   |
| <b>Study Goals</b>                     | <p>As a student, at the end of the course, I should be able to:</p> <ol style="list-style-type: none"> <li>1. Divide software development into a software development process life cycle according to the development methodologies (e.g., Scrum, Lean software development).</li> <li>2. Write software requirements</li> <li>3. Develop software following the software engineering practices</li> <li>4. Perform automated software testing</li> <li>5. Collaborate with software engineering and with other students from other faculties</li> </ol> |   |
| <b>Education Method</b>                | Lectures + readings + software project.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Literature and Study Materials</b>  | Slides and material on BrightSpace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Assessment</b>                      | <p>90% project + 10% presentation, grades have to be higher than 5.0.</p> <p>Repair<br/>The project will not have a repair opportunity at the end, but students will receive an evaluation on a preliminary report.</p> <p>Groups that received in between 4 and 5.7 for the presentation can redo the presentation with the limitation of a maximum grade of a 6.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)</p>                                                              |   |
| <b>Enrolment / Application</b>         | This course is only accessible to students who have registered for the Computer Science minor programme (via the regular registration procedure).                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Co-Instructor</b>                   | Prof.dr. A.E. Zaidman                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |

| WB3700                                 | Robotics Minor Project                                                                                                                                                                                                                           | 15 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <b>Responsible Instructor</b>          | M. Klomp                                                                                                                                                                                                                                         |    |
| <b>Contact Hours / Week</b><br>x/x/x/x | x/x/0/0                                                                                                                                                                                                                                          |    |
| <b>Education Period</b>                | 1<br>2                                                                                                                                                                                                                                           |    |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                |    |
| <b>Exam Period</b>                     | Different, to be announced                                                                                                                                                                                                                       |    |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                          |    |
| <b>Course Contents</b>                 | Building a robotic prototype for a real customer. This is the final project of the robotics minor. This is where all their learned objectives from the other courses in the minor should be visible and they should show what they have learned. |    |
| <b>Study Goals</b>                     | <p>After completing this course students should be able to:</p> <ul style="list-style-type: none"> <li>- work in a multidisciplinary team</li> <li>- contribute, from their own discipline, to a working robotic prototype</li> </ul>            |    |
| <b>Education Method</b>                | The students will be coached in working together in a multidisciplinary team, contact with customers, technological problems.                                                                                                                    |    |
| <b>Assessment</b>                      | In this course you will have multiple deliverables: a 'Detailed Design'-report (30%), the robotic prototype (40%), a 'Handover' (20%) and a peer review of group members (10%).                                                                  |    |
| <b>Department</b>                      | 3mE Department Cognitive Robotics                                                                                                                                                                                                                |    |

| WB3703                                 | Introduction to ROS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | M. Klomp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | x/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Exam Period</b>                     | Different, to be announced                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Contents</b>                 | <p>An introduction to the Robot Operating System (ROS). This course consists of a couple of lectures (possibly online) followed by an assignment on a small mobile robot.</p> <ul style="list-style-type: none"> <li>&gt; How to use ROS communication tools (topics, services, actions) to exchange information between functional modules</li> <li>&gt; Visualization and creation of a custom environment with a robot</li> <li>&gt; Mapping of the robot environment and navigation with a mobile robot</li> <li>&gt; How to implement a pick-and-place function with industrial robot arms</li> <li>&gt; Design of a complete robotic application with state machines</li> </ul> |   |
| <b>Study Goals</b>                     | <p>After completing this course students should be able to:</p> <ul style="list-style-type: none"> <li>- list the core concepts of the ROS computation graph</li> <li>- use the ROS filesystem</li> <li>- install and use existing ROS packages</li> <li>- create a custom ROS package</li> <li>- control a mobile robot</li> <li>- control an industrial robot</li> </ul>                                                                                                                                                                                                                                                                                                            |   |
| <b>Education Method</b>                | Lectures, followed by team assignment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Assessment</b>                      | A final assignment will have to be handed in. This is a demo on a physical robot. The assignment will be done in pairs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Department</b>                      | 3mE Department Cognitive Robotics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |

| WB3705                                 | Dynamics for Minor Robotics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 4 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | G. Papaioannou                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Instructor</b>                      | Ir. A.M. van der Niet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0.0.0.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Exam Period</b>                     | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Course Contents</b>                 | <p>Hibbeler Dynamics chapters;</p> <p>12: kinematics of a particle</p> <p>13: Force and acceleration ( particle )</p> <p>14: work and energy (particle)</p> <p>15: impulse and momentum (particle)</p> <p>16: planar kinematics (rigid body)</p> <p>17: force and acceleration (rigid body, planar)</p> <p>18: work and energy (rigid body, planar)</p> <p>19: impulse and momentum (rigid body, planar)</p> <p>22: vibrations</p>                                                                                                                                                                                                                                      |   |
| <b>Study Goals</b>                     | <p>LO1. Apply kinematics equation and mathematical formulations (position, velocity, acceleration notation) for the motion of point mass and of a rigid body</p> <p>LO2. Apply energy principles for the motion of point masses and rigid bodies.</p> <p>LO3. Formulate and solve the equations of motion for point masses and rigid bodies in different reference frames using vector notation.</p> <p>LO4. Apply the concepts of linear and angular momentum for the motion of rigid bodies in situations where impulse and collisions play a role.</p> <p>LO5. Analyse the motion of rigid bodies and the energy principles from data provided by lab exercises.</p> |   |
| <b>Education Method</b>                | <p>flipped classroom; watch lecture videos at home, instructional lectures at campus (not mandatory).</p> <p>Mandatory: 4 practice assignments, two lab reports</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Literature and Study Materials</b>  | Hibbeler Dynamics 14th edition SI units                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Assessment</b>                      | No written exam; Grading based on practice assignments and lab reports.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Department</b>                      | 3mE Department Cognitive Robotics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |

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|---------------------|-------------------------------------------------------|
| <b>Year</b>         | <b>2023/2024</b>                                      |
| <b>Organization</b> | <b>Mechanical, Maritime and Materials Engineering</b> |
| <b>Education</b>    | <b>Minors Mechanical Engineering</b>                  |

**Voor IO Bachelors**



| IO3881-21                              | Design in Robotics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Course Coordinator</b>              | Dr.ing. M.C. Rozendaal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Contact Hours / Week</b><br>x/x/x/x |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Contents</b>                 | In this IDE module, which is embedded within the Minor Robotics, we have bundled core IDE competencies within a coherent teaching program. The module teaches students how human and business aspects relate to and integrate with technology development. An iterative design methodology, inspired by SCRUM and the Basic Design Cycle (BDC), will be followed to foster this integration. This Module provides students with a parallel track (that will continue throughout the semester) to help them learn to apply human and business aspects in their project, across the different stages of technology development (i.e., from the fuzzy front-end to optimization).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Study Goals</b>                     | <p>Students will learn how technology is used and experienced and by people and will learn how to position technology development in a business context. Further, students will be spurred on to adopt a critical perspective on technology. During the Minor these aspects are directly linked their prototyping activities that they will engage with throughout the Minor. Concerning methodology, students will learn about iterative design processes (based on SCRUM), and will be trained in how to communicate with others during this process and to collaborate in multidisciplinary setting that the Minor Robotics provides. Appropriate methods and tools from the Delft Design Guide (DDG) will be provided to students as a resource.</p> <p>We have developed 6 different tracks that will be explained in more detail below:</p> <ol style="list-style-type: none"> <li>1. Human track: Understanding how technology is used and experienced by people. Contextual Inquiry, socio-technical systems, user-system interaction, co-design, user testing.</li> <li>2. Business track: Understanding technology development in a business context. Value proposition, unique selling point, multistakeholder analysis, business modeling, entrepreneurship.</li> <li>3. Prototyping track: Understanding prototyping as a vehicle for human, business technology integration. Experience prototyping, low fi-modeling (foam), Wizard of Oz approaches, interaction design/formgiving.</li> <li>4. Critical thinking track: Understanding different viewpoints and values. Problem (re)framing, philosophy of technology, ethics.</li> <li>5. Collaboration &amp; Communication track: Understanding the importance of multidisciplinary teamwork. Teambuilding, presentation and communication, multidisciplinary collaboration, visualization.</li> <li>6. Design Process track: Understanding iterative design processes. SCRUM methodology, Basic Design Cycle</li> </ol> |   |
| <b>Education Method</b>                | This Module provides a parallel track in the Minor program. At different stages of their technology development, IDE knowledge and skills are provided that are appropriate and useful to them at that specific stage. The modules are organized in SCRUM sprints that are blocks of one or two weeks, in which a basic design cycle is followed. In each sprint a specific assignment is given to the students, that provides them with specific IDE knowledge and skills that they can directly apply.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Literature and Study Materials</b>  | Delft Design Guide, van Boeijen et al.<br>Integraal Product Ontwerpen, Roozenburg en Eekels                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Assessment</b>                      | The means in which the students learning is assessed differs between courses. Assessment is done by reviewing the design outcomes/deliverables (prototypes, individual reports, personal reflections) both formative and summative.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Special Information</b>             | Dr. ing. Marco Rozendaal, responsible teacher ID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |

| TI3105TU                                | Intermediate Python Programming                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 5 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | F. Mulder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 4/0/0/0 lecture; 4/0/0/0 lab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Education Period</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Start Education</b>                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Exam Period</b>                      | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Expected prior knowledge</b>         | You are expected to have finished our Python Prerequisites mini-course before this course starts. Find it at <a href="https://ipp.pages.ewi.tudelft.nl/python-prerequisites/">https://ipp.pages.ewi.tudelft.nl/python-prerequisites/</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Contents</b>                  | This course teaches programming in Python on an intermediate level.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Study Goals</b>                      | <p>After completion of this course, the student is able to:</p> <ol style="list-style-type: none"> <li>1. Implement Python programs based on a given natural-language description.</li> <li>2. Write code that follows Python's code style guidelines.</li> <li>3. Make use of basic data structures such as strings, lists, dictionaries, tuples and sets.</li> <li>4. Apply object-oriented programming concepts such as inheritance, operator overloading and properties.</li> <li>5. Apply functional programming concepts such as list comprehensions, recursion and higher-order functions, as well as iterators and generators.</li> <li>6. Implement reading and writing data from/to different file formats.</li> <li>7. Understand and write type hints.</li> </ol>                                                                                                                                                                                                                       |   |
| <b>Education Method</b>                 | Seminars and lab sessions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Computer Use</b>                     | Students do all programming assignments on their own computers (except for the exams).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Books</b>                            | Think Python 2nd Edition by Allen B. Downey (freely available from <a href="http://thinkpython2.com">thinkpython2.com</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Assessment</b>                       | <p>Midterm exam (2 hours) in week 1.5, final exam (3 hours) in week 1.10. Both exams are computer exams. The midterm exam covers the course materials discussed until week 1.5, and the final exam covers all course materials.</p> <p>There is also a resit opportunity. The resit covers all course materials (there is no separate resit for the midterm exam).</p> <p>The final grade of the course is calculated as follows:</p> $\text{course grade} = 0.2 * \max(\text{midterm}, \text{final}) + 0.8 * \text{final}$ <p>So the midterm grade only counts if it is higher than the grade for the final exam.</p> <p>The final exam grade has to be at least 5.0. (And as usual, the course grade has to be at least 5.8 in order to pass the course.)</p> <p>The resit grade replaces the original grade only if the resit grade is greater than the original grade.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)</p> |   |
| <b>Permitted Materials during Tests</b> | You may use the Python documentation during the exam.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Enrolment / Application</b>          | This course is only accessible to students who have registered for one of these minor programmes (via the regular registration procedure): Computer Science, Engineering with AI or Robotics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Co-Instructor</b>                    | I.D. van Kreveld                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |

| WB2632                                 | Advanced Engineering Design Project                 | 6 |
|----------------------------------------|-----------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. A. van Beek                                  |   |
| <b>Instructor</b>                      | Dr.ir. R.A.J. van Ostayen                           |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 8.0.0.0                                             |   |
| <b>Education Period</b>                | 1                                                   |   |
| <b>Start Education</b>                 | 1                                                   |   |
| <b>Exam Period</b>                     | Different, to be announced                          |   |
| <b>Course Language</b>                 | Dutch                                               |   |
| <b>Remarks</b>                         | Hoorcollege online 4/0/0/0<br>Projecturen 4/0/0/0   |   |
| <b>Department</b>                      | 3mE Department Precision & Microsystems Engineering |   |

| WB3700                                 | Robotics Minor Project                                                                                                                                                                                                                           | 15 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <b>Responsible Instructor</b>          | M. Klomp                                                                                                                                                                                                                                         |    |
| <b>Contact Hours / Week</b><br>x/x/x/x | x/x/0/0                                                                                                                                                                                                                                          |    |
| <b>Education Period</b>                | 1<br>2                                                                                                                                                                                                                                           |    |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                |    |
| <b>Exam Period</b>                     | Different, to be announced                                                                                                                                                                                                                       |    |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                          |    |
| <b>Course Contents</b>                 | Building a robotic prototype for a real customer. This is the final project of the robotics minor. This is where all their learned objectives from the other courses in the minor should be visible and they should show what they have learned. |    |
| <b>Study Goals</b>                     | After completing this course students should be able to:<br>- work in a multidisciplinary team<br>- contribute, from their own discipline, to a working robotic prototype                                                                        |    |
| <b>Education Method</b>                | The students will be coached in working together in a multidisciplinary team, contact with customers, technological problems.                                                                                                                    |    |
| <b>Assessment</b>                      | In this course you will have multiple deliverables: a 'Detailed Design'-report (30%), the robotic prototype (40%), a 'Handover' (20%) and a peer review of group members (10%).                                                                  |    |
| <b>Department</b>                      | 3mE Department Cognitive Robotics                                                                                                                                                                                                                |    |

|                     |                                                       |
|---------------------|-------------------------------------------------------|
| <b>Year</b>         | <b>2023/2024</b>                                      |
| <b>Organization</b> | <b>Mechanical, Maritime and Materials Engineering</b> |
| <b>Education</b>    | <b>Minors Mechanical Engineering</b>                  |

**Voor WB Bachelors**

| ET3033TU                                | Circuit Analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3 |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr.ir. N.P. van der Meijs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 3/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Period</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Start Education</b>                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Exam Period</b>                      | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Expected prior knowledge</b>         | The course requires familiarity with handling the complex numbers algebra.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Contents</b>                  | <p>Electrical circuits fundamentals / basic concepts and components: current and charge, voltage and current sources, electrical circuits, circuit symbols, direct current, alternating current, resistors capacitors and inductors, Ohm's law, Kirchhoff's laws, power dissipation in resistors, series and parallel connections.</p> <p>Measurement of voltages and currents: common wave forms, measuring techniques and equipment.</p> <p>Resistance and DC Circuits: Thévenin and Norton theorems, superposition, nodal analysis, mesh analysis, (solving) circuit equations.</p> <p>Capacitance and electric fields: capacitors and capacitance, electric field strength and flux density, series and parallel connected capacitors, capacitor I/V relationship, sinusoidal voltage and current, energy stored in a capacitor.</p> <p>Inductance and Magnetic Fields: electromagnetism, reluctance, inductance, self-inductance, series and parallel connections, inductor I/V relationship, sinusoidal voltage and current, energy stored in an inductor, mutual inductance, transformers.</p> <p>Alternating voltages and currents: resistance vs. reactance, impedance, phasor diagrams, complex notation.</p> <p>Power in AC circuits: power dissipation in resistive circuits, power in capacitors and inductors, power in circuits with resistance and reactance, active and reactive power, power factor correction, three phase systems, power measurement.</p> <p>Frequency characteristics of AC circuits: two-port networks, the decibel, frequency response, filter networks, Bode diagrams, RLC circuits and resonance, stray capacitance and inductance.</p> <p>Transient behaviour: Charging of capacitors and energising of inductors, discharging and de-energising, first order systems, generalised response, second order systems, higher order systems.</p> |   |
| <b>Study Goals</b>                      | The course aims at building up insight and analysis instruments in the area of linear electrical circuits and components. It will also establish a solid foundation for understanding, analysing and synthesising electronic circuits using non-linear active components in both the analogue and digital domains.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Method</b>                 | Interactive lecturing as a preparation for more detailed self-study and engagement with the content. Lectures are not necessarily completely covering all topics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Literature and Study Materials</b>   | <p>Neil Storey, "Electronics, a Systems Approach", 6th edition, Pearson Education, ISBN 978-0273773276.</p> <p>Material in the Brightspace learning environment.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Assessment</b>                       | Homeworks + written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Permitted Materials during Tests</b> | <p>The exam is a closed book exam. However, a single-sided self-handwritten A4 as a cheat sheet is permitted.</p> <p>Only a non-programmable calculator without capabilities for storage of text or formulas and without capabilities for graphing may be used. Suitable types are the TI-30 and the Casio FX-82, and the various variants of these machines.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Judgement</b>                        | The final grade depends for 10% on the homework and for 90% on the written exam.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>maximum aantal deelnemers</b>        | 100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |

| IO3881-21                              | Design in Robotics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Course Coordinator</b>              | Dr.ing. M.C. Rozendaal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Contact Hours / Week</b><br>x/x/x/x |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Contents</b>                 | In this IDE module, which is embedded within the Minor Robotics, we have bundled core IDE competencies within a coherent teaching program. The module teaches students how human and business aspects relate to and integrate with technology development. An iterative design methodology, inspired by SCRUM and the Basic Design Cycle (BDC), will be followed to foster this integration. This Module provides students with a parallel track (that will continue throughout the semester) to help them learn to apply human and business aspects in their project, across the different stages of technology development (i.e., from the fuzzy front-end to optimization).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Study Goals</b>                     | <p>Students will learn how technology is used and experienced and by people and will learn how to position technology development in a business context. Further, students will be spurred on to adopt a critical perspective on technology. During the Minor these aspects are directly linked their prototyping activities that they will engage with throughout the Minor. Concerning methodology, students will learn about iterative design processes (based on SCRUM), and will be trained in how to communicate with others during this process and to collaborate in multidisciplinary setting that the Minor Robotics provides. Appropriate methods and tools from the Delft Design Guide (DDG) will be provided to students as a resource.</p> <p>We have developed 6 different tracks that will be explained in more detail below:</p> <ol style="list-style-type: none"> <li>1. Human track: Understanding how technology is used and experienced by people. Contextual Inquiry, socio-technical systems, user-system interaction, co-design, user testing.</li> <li>2. Business track: Understanding technology development in a business context. Value proposition, unique selling point, multistakeholder analysis, business modeling, entrepreneurship.</li> <li>3. Prototyping track: Understanding prototyping as a vehicle for human, business technology integration. Experience prototyping, low fi-modeling (foam), Wizard of Oz approaches, interaction design/formgiving.</li> <li>4. Critical thinking track: Understanding different viewpoints and values. Problem (re)framing, philosophy of technology, ethics.</li> <li>5. Collaboration &amp; Communication track: Understanding the importance of multidisciplinary teamwork. Teambuilding, presentation and communication, multidisciplinary collaboration, visualization.</li> <li>6. Design Process track: Understanding iterative design processes. SCRUM methodology, Basic Design Cycle</li> </ol> |   |
| <b>Education Method</b>                | This Module provides a parallel track in the Minor program. At different stages of their technology development, IDE knowledge and skills are provided that are appropriate and useful to them at that specific stage. The modules are organized in SCRUM sprints that are blocks of one or two weeks, in which a basic design cycle is followed. In each sprint a specific assignment is given to the students, that provides them with specific IDE knowledge and skills that they can directly apply.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Literature and Study Materials</b>  | Delft Design Guide, van Boeijen et al.<br>Integraal Product Ontwerpen, Roozenburg en Eekels                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Assessment</b>                      | The means in which the students learning is assessed differs between courses. Assessment is done by reviewing the design outcomes/deliverables (prototypes, individual reports, personal reflections) both formative and summative.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Special Information</b>             | Dr. ing. Marco Rozendaal, responsible teacher ID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |

| TI3115TU                               | Software Engineering Methods                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 5 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | T. Durieux                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/0/0 lecture; 4/0/0/0 lab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Exam Period</b>                     | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Expected prior knowledge</b>        | Basics of (Python) programming or currently taking TI3105TU.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Contents</b>                 | <p>In this course, students will learn:</p> <ul style="list-style-type: none"> <li>- software development processes, such as waterfall and agile</li> <li>- requirements engineering (use cases + user stories)</li> <li>- version control and Git</li> <li>- software testing</li> <li>- software design and design patterns</li> <li>- refactoring, code smells, and implementation best practices</li> </ul>                                                                                                                                          |   |
| <b>Study Goals</b>                     | <p>As a student, at the end of the course, I should be able to:</p> <ol style="list-style-type: none"> <li>1. Divide software development into a software development process life cycle according to the development methodologies (e.g., Scrum, Lean software development).</li> <li>2. Write software requirements</li> <li>3. Develop software following the software engineering practices</li> <li>4. Perform automated software testing</li> <li>5. Collaborate with software engineering and with other students from other faculties</li> </ol> |   |
| <b>Education Method</b>                | Lectures + readings + software project.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Literature and Study Materials</b>  | Slides and material on BrightSpace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Assessment</b>                      | <p>90% project + 10% presentation, grades have to be higher than 5.0.</p> <p>Repair<br/>The project will not have a repair opportunity at the end, but students will receive an evaluation on a preliminary report.</p> <p>Groups that received in between 4 and 5.7 for the presentation can redo the presentation with the limitation of a maximum grade of a 6.</p> <p>Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)</p>                                                              |   |
| <b>Enrolment / Application</b>         | This course is only accessible to students who have registered for the Computer Science minor programme (via the regular registration procedure).                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Co-Instructor</b>                   | Prof.dr. A.E. Zaidman                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |

| WB3168                                 | Robotic System Identification and Parameter Estimation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Ir. C.C. Claij                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/1/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | Different, to be announced                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Contents</b>                 | <p>Selected topics on sensors for robotics<br/>The course consists of 1-2 theoretical and 1-2 practical lectures.</p> <p>The theoretical lecture(s) focus on gaining knowledge on different topics and sensors:</p> <ol style="list-style-type: none"> <li>a. Introduction to sensors (classification and characteristics)</li> <li>b. Visual sensors (2D Cameras)</li> <li>c. Depth sensors (Time-of-Flight)</li> <li>d. Proximity sensors (IR and ultrasonic)</li> <li>e. Beacon-based sensors (RFID and GPS)</li> <li>f. Position sensors (Encoders)</li> <li>g. Orientation sensors (Compass and IMU)</li> <li>h. Force/Torque sensors</li> <li>i. Pressure sensors</li> <li>j. Temperature sensors</li> <li>k. Data processing</li> <li>l. Calibration</li> </ol> <p>The practical lecture(s) aim on experiencing the knowledge gained during the theoretical lectures.</p> |   |
| <b>Study Goals</b>                     | <p>This course aims to provide an overview of sensors and their features for robotic applications. Next to the theoretical understanding of the use of different sensors, you will experience several of their use cases and limitations.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Assessment</b>                      | Written report                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Department</b>                      | 3mE Department Cognitive Robotics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |

| WB3700                                 | Robotics Minor Project                                                                                                                                                                                                                           | 15 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <b>Responsible Instructor</b>          | M. Klomp                                                                                                                                                                                                                                         |    |
| <b>Contact Hours / Week</b><br>x/x/x/x | x/x/0/0                                                                                                                                                                                                                                          |    |
| <b>Education Period</b>                | 1<br>2                                                                                                                                                                                                                                           |    |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                |    |
| <b>Exam Period</b>                     | Different, to be announced                                                                                                                                                                                                                       |    |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                          |    |
| <b>Course Contents</b>                 | Building a robotic prototype for a real customer. This is the final project of the robotics minor. This is where all their learned objectives from the other courses in the minor should be visible and they should show what they have learned. |    |
| <b>Study Goals</b>                     | After completing this course students should be able to:<br>- work in a multidisciplinary team<br>- contribute, from their own discipline, to a working robotic prototype                                                                        |    |
| <b>Education Method</b>                | The students will be coached in working together in a multidisciplinary team, contact with customers, technological problems.                                                                                                                    |    |
| <b>Assessment</b>                      | In this course you will have multiple deliverables: a 'Detailed Design'-report (30%), the robotic prototype (40%), a 'Handover' (20%) and a peer review of group members (10%).                                                                  |    |
| <b>Department</b>                      | 3mE Department Cognitive Robotics                                                                                                                                                                                                                |    |

| WB3703                                 | Introduction to ROS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | M. Klomp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | x/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Exam Period</b>                     | Different, to be announced                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Contents</b>                 | An introduction to the Robot Operating System (ROS). This course consists of a couple of lectures (possibly online) followed by an assignment on a small mobile robot.<br><br>> How to use ROS communication tools (topics, services, actions) to exchange information between functional modules<br>> Visualization and creation of a custom environment with a robot<br>> Mapping of the robot environment and navigation with a mobile robot<br>> How to implement a pick-and-place function with industrial robot arms<br>> Design of a complete robotic application with state machines |   |
| <b>Study Goals</b>                     | After completing this course students should be able to:<br>- list the core concepts of the ROS computation graph<br>- use the ROS filesystem<br>- install and use existing ROS packages<br>- create a custom ROS package<br>- control a mobile robot<br>- control an industrial robot                                                                                                                                                                                                                                                                                                       |   |
| <b>Education Method</b>                | Lectures, followed by team assignment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Assessment</b>                      | A final assignment will have to be handed in. This is a demo on a physical robot. The assignment will be done in pairs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Department</b>                      | 3mE Department Cognitive Robotics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |



|                     |                                                       |
|---------------------|-------------------------------------------------------|
| <b>Year</b>         | <b>2023/2024</b>                                      |
| <b>Organization</b> | <b>Mechanical, Maritime and Materials Engineering</b> |
| <b>Education</b>    | <b>Minors Mechanical Engineering</b>                  |

**Voor TI Bachelors**

| ET3033TU                                | Circuit Analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3 |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr.ir. N.P. van der Meijs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 3/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Education Period</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Start Education</b>                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Exam Period</b>                      | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Expected prior knowledge</b>         | The course requires familiarity with handling the complex numbers algebra.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Contents</b>                  | <p>Electrical circuits fundamentals / basic concepts and components: current and charge, voltage and current sources, electrical circuits, circuit symbols, direct current, alternating current, resistors capacitors and inductors, Ohm's law, Kirchhoff's laws, power dissipation in resistors, series and parallel connections.</p> <p>Measurement of voltages and currents: common wave forms, measuring techniques and equipment.</p> <p>Resistance and DC Circuits: Thévenin and Norton theorems, superposition, nodal analysis, mesh analysis, (solving) circuit equations.</p> <p>Capacitance and electric fields: capacitors and capacitance, electric field strength and flux density, series and parallel connected capacitors, capacitor I/V relationship, sinusoidal voltage and current, energy stored in a capacitor.</p> <p>Inductance and Magnetic Fields: electromagnetism, reluctance, inductance, self-inductance, series and parallel connections, inductor I/V relationship, sinusoidal voltage and current, energy stored in an inductor, mutual inductance, transformers.</p> <p>Alternating voltages and currents: resistance vs. reactance, impedance, phasor diagrams, complex notation.</p> <p>Power in AC circuits: power dissipation in resistive circuits, power in capacitors and inductors, power in circuits with resistance and reactance, active and reactive power, power factor correction, three phase systems, power measurement.</p> <p>Frequency characteristics of AC circuits: two-port networks, the decibel, frequency response, filter networks, Bode diagrams, RLC circuits and resonance, stray capacitance and inductance.</p> <p>Transient behaviour: Charging of capacitors and energising of inductors, discharging and de-energising, first order systems, generalised response, second order systems, higher order systems.</p> |   |
| <b>Study Goals</b>                      | The course aims at building up insight and analysis instruments in the area of linear electrical circuits and components. It will also establish a solid foundation for understanding, analysing and synthesising electronic circuits using non-linear active components in both the analogue and digital domains.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Method</b>                 | Interactive lecturing as a preparation for more detailed self-study and engagement with the content. Lectures are not necessarily completely covering all topics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Literature and Study Materials</b>   | <p>Neil Storey, "Electronics, a Systems Approach", 6th edition, Pearson Education, ISBN 978-0273773276.</p> <p>Material in the Brightspace learning environment.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Assessment</b>                       | Homeworks + written exam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Permitted Materials during Tests</b> | <p>The exam is a closed book exam. However, a single-sided self-handwritten A4 as a cheat sheet is permitted.</p> <p>Only a non-programmable calculator without capabilities for storage of text or formulas and without capabilities for graphing may be used. Suitable types are the TI-30 and the Casio FX-82, and the various variants of these machines.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Judgement</b>                        | The final grade depends for 10% on the homework and for 90% on the written exam.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>maximum aantal deelnemers</b>        | 100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |

| IO3881-21                              | Design in Robotics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Course Coordinator</b>              | Dr.ing. M.C. Rozendaal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Contact Hours / Week</b><br>x/x/x/x |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Education Period</b>                | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Contents</b>                 | In this IDE module, which is embedded within the Minor Robotics, we have bundled core IDE competencies within a coherent teaching program. The module teaches students how human and business aspects relate to and integrate with technology development. An iterative design methodology, inspired by SCRUM and the Basic Design Cycle (BDC), will be followed to foster this integration. This Module provides students with a parallel track (that will continue throughout the semester) to help them learn to apply human and business aspects in their project, across the different stages of technology development (i.e., from the fuzzy front-end to optimization).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Study Goals</b>                     | <p>Students will learn how technology is used and experienced and by people and will learn how to position technology development in a business context. Further, students will be spurred on to adopt a critical perspective on technology. During the Minor these aspects are directly linked their prototyping activities that they will engage with throughout the Minor. Concerning methodology, students will learn about iterative design processes (based on SCRUM), and will be trained in how to communicate with others during this process and to collaborate in multidisciplinary setting that the Minor Robotics provides. Appropriate methods and tools from the Delft Design Guide (DDG) will be provided to students as a resource.</p> <p>We have developed 6 different tracks that will be explained in more detail below:</p> <ol style="list-style-type: none"> <li>1. Human track: Understanding how technology is used and experienced by people. Contextual Inquiry, socio-technical systems, user-system interaction, co-design, user testing.</li> <li>2. Business track: Understanding technology development in a business context. Value proposition, unique selling point, multistakeholder analysis, business modeling, entrepreneurship.</li> <li>3. Prototyping track: Understanding prototyping as a vehicle for human, business technology integration. Experience prototyping, low fi-modeling (foam), Wizard of Oz approaches, interaction design/formgiving.</li> <li>4. Critical thinking track: Understanding different viewpoints and values. Problem (re)framing, philosophy of technology, ethics.</li> <li>5. Collaboration &amp; Communication track: Understanding the importance of multidisciplinary teamwork. Teambuilding, presentation and communication, multidisciplinary collaboration, visualization.</li> <li>6. Design Process track: Understanding iterative design processes. SCRUM methodology, Basic Design Cycle</li> </ol> |   |
| <b>Education Method</b>                | This Module provides a parallel track in the Minor program. At different stages of their technology development, IDE knowledge and skills are provided that are appropriate and useful to them at that specific stage. The modules are organized in SCRUM sprints that are blocks of one or two weeks, in which a basic design cycle is followed. In each sprint a specific assignment is given to the students, that provides them with specific IDE knowledge and skills that they can directly apply.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Literature and Study Materials</b>  | Delft Design Guide, van Boeijen et al.<br>Integraal Product Ontwerpen, Roozenburg en Eekels                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Assessment</b>                      | The means in which the students learning is assessed differs between courses. Assessment is done by reviewing the design outcomes/deliverables (prototypes, individual reports, personal reflections) both formative and summative.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Special Information</b>             | Dr. ing. Marco Rozendaal, responsible teacher ID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |

| WB3700                                 | Robotics Minor Project                                                                                                                                                                                                                           | 15 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <b>Responsible Instructor</b>          | M. Klomp                                                                                                                                                                                                                                         |    |
| <b>Contact Hours / Week</b><br>x/x/x/x | x/x/0/0                                                                                                                                                                                                                                          |    |
| <b>Education Period</b>                | 1<br>2                                                                                                                                                                                                                                           |    |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                |    |
| <b>Exam Period</b>                     | Different, to be announced                                                                                                                                                                                                                       |    |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                          |    |
| <b>Course Contents</b>                 | Building a robotic prototype for a real customer. This is the final project of the robotics minor. This is where all their learned objectives from the other courses in the minor should be visible and they should show what they have learned. |    |
| <b>Study Goals</b>                     | <p>After completing this course students should be able to:</p> <ul style="list-style-type: none"> <li>- work in a multidisciplinary team</li> <li>- contribute, from their own discipline, to a working robotic prototype</li> </ul>            |    |
| <b>Education Method</b>                | The students will be coached in working together in a multidisciplinary team, contact with customers, technological problems.                                                                                                                    |    |
| <b>Assessment</b>                      | In this course you will have multiple deliverables: a 'Detailed Design'-report (30%), the robotic prototype (40%), a 'Handover' (20%) and a peer review of group members (10%).                                                                  |    |
| <b>Department</b>                      | 3mE Department Cognitive Robotics                                                                                                                                                                                                                |    |

| WB3703                                 | Introduction to ROS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | M. Klomp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | x/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Exam Period</b>                     | Different, to be announced                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Course Contents</b>                 | <p>An introduction to the Robot Operating System (ROS). This course consists of a couple of lectures (possibly online) followed by an assignment on a small mobile robot.</p> <ul style="list-style-type: none"> <li>&gt; How to use ROS communication tools (topics, services, actions) to exchange information between functional modules</li> <li>&gt; Visualization and creation of a custom environment with a robot</li> <li>&gt; Mapping of the robot environment and navigation with a mobile robot</li> <li>&gt; How to implement a pick-and-place function with industrial robot arms</li> <li>&gt; Design of a complete robotic application with state machines</li> </ul> |   |
| <b>Study Goals</b>                     | <p>After completing this course students should be able to:</p> <ul style="list-style-type: none"> <li>- list the core concepts of the ROS computation graph</li> <li>- use the ROS filesystem</li> <li>- install and use existing ROS packages</li> <li>- create a custom ROS package</li> <li>- control a mobile robot</li> <li>- control an industrial robot</li> </ul>                                                                                                                                                                                                                                                                                                            |   |
| <b>Education Method</b>                | Lectures, followed by team assignment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Assessment</b>                      | A final assignment will have to be handed in. This is a demo on a physical robot. The assignment will be done in pairs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Department</b>                      | 3mE Department Cognitive Robotics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |

| WB3704                                 | Gazebo Assignment                                                                                                                                                                                                                                           | 2 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | M. Klomp                                                                                                                                                                                                                                                    |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | x/0/0/0                                                                                                                                                                                                                                                     |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                           |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                           |   |
| <b>Exam Period</b>                     | Different, to be announced                                                                                                                                                                                                                                  |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                     |   |
| <b>Course Contents</b>                 | <p>In this course the student will learn about the use of the Gazebo simulator in ROS. Firstly creating your own model of the robot, and then controlling the robot, reading out sensor values, and interacting with the environment.</p>                   |   |
| <b>Study Goals</b>                     | <p>After this course the student will be able to:</p> <ul style="list-style-type: none"> <li>- Understand the importance of a simulation of the robot.</li> <li>- Build a working simulation of an actual robot including sensors and actuators.</li> </ul> |   |
| <b>Education Method</b>                | One 45 min lecture, followed by an assignment.                                                                                                                                                                                                              |   |
| <b>Assessment</b>                      | The final assignment of this course will be a working Gazebo simulation of a to be designed robot.                                                                                                                                                          |   |
| <b>Department</b>                      | 3mE Department Cognitive Robotics                                                                                                                                                                                                                           |   |

| WB3705                                 | Dynamics for Minor Robotics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 4 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | G. Papaioannou                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Instructor</b>                      | Ir. A.M. van der Niet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0.0.0.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Exam Period</b>                     | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Contents</b>                 | Hibbeler Dynamics chapters;<br>12: kinematics of a particle<br>13: Force and acceleration ( particle )<br>14: work and energy (particle)<br>15: impulse and momentum (particle)<br>16: planar kinematics (rigid body)<br>17: force and acceleration (rigid body, planar)<br>18: work and energy (rigid body, planar)<br>19: impulse and momentum (rigid body, planar)<br>22: vibrations                                                                                                                                                                                                                                                          |   |
| <b>Study Goals</b>                     | LO1. Apply kinematics equation and mathematical formulations (position, velocity, acceleration notation) for the motion of point mass and of a rigid body<br>LO2. Apply energy principles for the motion of point masses and rigid bodies.<br>LO3. Formulate and solve the equations of motion for point masses and rigid bodies in different reference frames using vector notation.<br>LO4. Apply the concepts of linear and angular momentum for the motion of rigid bodies in situations where impulse and collisions play a role.<br>LO5. Analyse the motion of rigid bodies and the energy principles from data provided by lab exercises. |   |
| <b>Education Method</b>                | flipped classroom; watch lecture videos at home, instructional lectures at campus (not mandatory).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
|                                        | Mandatory: 4 practice assignments, two lab reports                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Literature and Study Materials</b>  | Hibbeler Dynamics 14th edition SI units                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                      | No written exam; Grading based on practice assignments and lab reports.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Department</b>                      | 3mE Department Cognitive Robotics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |

|                     |                                                       |
|---------------------|-------------------------------------------------------|
| <b>Year</b>         | <b>2023/2024</b>                                      |
| <b>Organization</b> | <b>Mechanical, Maritime and Materials Engineering</b> |
| <b>Education</b>    | <b>Minors Mechanical Engineering</b>                  |

## WB-MI-217-23

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|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Contact for students</b>                   | Prof.dr.ir. Wiebren de Jong<br>Section Large-Scale Energy storage (LSE)<br>Faculty 3mE - Department of Process & Energy<br>Leeghwaterstraag 39, office 34 J-0-920 (lab of Process & Energy)<br>Telephone: 015-2789476<br>Mobile phone: 06-51236425<br>Mail: wiebren.dejong@tudelft.nl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Program Title</b>                          | ELECS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Director of Education</b>                  | Dr. R. Delfos<br>Faculty 3mE<br>r.delfos@tudelft.nl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>EC Program</b>                             | 30 ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Program Start (Study Year)</b>             | 2019                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Introduction 1</b>                         | Energy conversion is the engine of the economics of our society. For a long time since the industrial revolution in the 18th century fossil fuels have dominated energy supply to a multitude of industries, the transportation sector and households. Reduction of CO2 emissions is urgently needed in view of Climate Change. Therefore, alternative energy supply is coming up rapidly. More and more, electricity will form the energy carrier for processes and household heat and power supply. Solar and wind power supply are increasingly implemented and become cheaper in fast pace. However, solar and wind are intermittent in nature and their power supply characteristics do not match the demand, neither in time nor place. There are periods and places with excess electricity supply compared to the demand, but certainly also times and locations with power supply shortage. Energy storage therefore is needed. In order to seriously reduce our societies' dependency on fossil fuels a huge effort is currently made to develop, scale-up and implement energy storage systems. Such systems are required for both short term (daily fluctuations and day-night rhythm) and long term (bridging reliable energy supply over seasons). This minor 'ELECS' covers both the aspects of renewable energy supply, energy conversion and storage and system integration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Program Goals</b>                          | <p>The aim of the minor is to equip bachelor engineering students with selected knowledge and skills concerning energy conversion and storage systems and component technologies that are based on renewable resources, which are as such mostly fluctuating and not-well predictable. Electricity is not easily and cost-effectively stored, so a mismatch both in time and geographically-between demand and supply of power is a real societal challenge. Also, storage of electricity needs to be accomplished at widely varying time scales to cope with fast weather fluctuations, day-night rhythm, occurrence of larger weather systems effects and seasonal variation of supply.</p> <p>This minor will deal with providing the fundamentals of a the most relevant intermittent renewable energy supply systems, wind and solar energy systems. Furthermore, systems analysis performance evaluation courses (energy storage, life cycle assessment) and practical studies (system modelling of energy conversion and storage systems and component level study design/research project) will be offered to enable students to make an evaluation of a large renewables based energy conversion and storage system.</p> <p>The difference with other existing minors that are offered within TU Delft is that a group of renewable energy systems is considered here in the framework of the challenge of storing energy at different time scales, taking a systems evaluation and practical component study approach together, which is unique. In summary, the topic of the minor is highly relevant due to the energy transition the world faces. Because the minor provides a comprehensive introduction to renewable energies, providing fundamental and practical knowledge and skills to students in this domain, it will be attractive and challenging to our engineering students.</p> |
| <b>Administration by the Education of</b>     | Faculty 3mE<br>Mechanical Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Minor Title</b>                            | Engineering for Large-scale energy conversion and storage ELECS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>EC Minor (Volume)</b>                      | 30 ECTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Intended for</b>                           | Targeted students for the Minor ELECS are those with an engineering basic knowledge of thermodynamics, heat transfer and/or fluid mechanics; i.e. students of Mechanical Engineering, Aerospace Engineering, Applied Sciences, Electrical Engineering, Technology Policy and Management, and Civil Engineering (Strictly: MT, Wb, CT, EE, TA, AE, TB, LST, MST, NB, TN)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Minor Content 1</b>                        | <p>The minor ELECS will provide students with an attractive constellation of curriculum courses and a multi-disciplinary integrated project concerning the topic of energy conversion and storage engineering. Aspects of variability in power (and heat) production characteristics and demand matching issues will be dealt with, combined with the evaluation of conversion characteristics and energy system evaluation. The undergraduate student having followed this Minor will be able to:</p> <ul style="list-style-type: none"> <li>- Understand and apply knowledge of renewable power and heat supply systems</li> <li>- Apply process calculations for energy conversion and storage systems based on thermodynamics, mass and heat transfer and fluid mechanics</li> <li>- Evaluate systems based on the concepts of exergy analysis and life cycle assessment</li> <li>- Provide insights into current state of the art and future trends in energy engineering</li> <li>- Synthesize their knowledge and competences in a final design project, including aspects of technical communication (presentations, technical report writing).</li> </ul> <p>Targeted students for the Minor ELECS are those with an engineering basic knowledge of thermodynamics, heat transfer and/or fluid mechanics; i.e. students of Mechanical Engineering, Aerospace Engineering, Applied Sciences, Electrical Engineering, Technology Policy and Management, and Civil Engineering (Strictly: MT, Wb, CT, EE, TA, AE, TB, LST, MST, NB, TN)</p>                                                                                                                                                                                                                                                                                                                                                         |
| <b>Remarks Maximum number of participants</b> | 75                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

| AE3516A                                | Fundamentals of Wind Energy I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 3 |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. A.C. Viré                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Instructor</b>                      | Dr. D.A. von Terzi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 2/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Exam Period</b>                     | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Course Contents</b>                 | <p>This course introduces the topic of offshore wind energy by going through the main steps that a wind farm developer needs to achieve in order to build an offshore wind farm. The main topics cover in the course are:</p> <ul style="list-style-type: none"> <li>- Historical developments in wind energy</li> <li>- Wind energy harvesting and Betz limit</li> <li>- Offshore wind and wave conditions</li> <li>- Estimation of the energy yield</li> <li>- Design considerations for wind turbine farms</li> <li>- Economic aspects</li> <li>- Planning issues</li> <li>- Emerging concepts such as floating wind turbines</li> </ul>                                                                |   |
| <b>Study Goals</b>                     | <p>At the end of this course, the students will be able to:</p> <ul style="list-style-type: none"> <li>- Explain the basic principles of wind energy conversion.</li> <li>- Identify the steps required when developing offshore wind farms.</li> <li>- Differentiate between the challenges and opportunities of offshore and onshore wind energy harvesting.</li> <li>- Use existing tools for estimating the power produced by an offshore wind farm and the effect of wakes.</li> <li>- Integrate knowledge from various fields of engineering related to the development of offshore wind parks.</li> <li>- Reflect on the current and future trends in the field of offshore wind energy.</li> </ul> |   |
| <b>Education Method</b>                | <p>The teaching method uses the principle of flipped classroom. This means that, every week, the students have to perform online assignments (e.g. watch a lecture, read a text, etc) prior to coming to class. In class, additional lecture materials, exercises and feedback are given to the students.</p>                                                                                                                                                                                                                                                                                                                                                                                              |   |
| <b>Assessment</b>                      | <p>The final assessment is a written exam. Additional points are given for performing the online activities prior to class.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |

| ET3034TU                               | Solar Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Prof.dr.ir. A.H.M. Smets                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/2/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Contents</b>                 | <p>In the course Solar Energy, you will learn to design a complete photovoltaic (PV) system for any application and location. This course introduces the technology that converts solar energy into electricity. The role of solar energy in both the energy transition towards a sustainable future and climate change mitigation will be discussed in detail.</p> <p>The physical principle of the PV energy conversion together with the design rules for a solar cell are introduced. These design rules are applied on the various PV technologies from cell up to module level. This program highlights various metrics related to performance, costs, reliability and circularity that will be used to evaluate the advantages and limitation of different PV technologies and its applications.</p> |   |
| <b>Study Goals</b>                     | <p>After completion of the course, the students are able to:</p> <ul style="list-style-type: none"> <li>a) Reflect on the role of solar energy in both the energy transition towards a sustainable future and climate change mitigation</li> <li>b) Understand the physical working principles of photovoltaic energy conversion in solar cells</li> <li>c) Recognize and describe the various photovoltaic technologies and evaluate their applicability using various metrics related to performance, costs, reliability and circularity of PV.</li> <li>d) Design a complete photovoltaic system for any particular application and location</li> </ul>                                                                                                                                                  |   |
| <b>Education Method</b>                | <p>The education method is the so-called flipped-class room.</p> <p>Lectures will be provided in small video lectures on Brightspace.</p> <p>The on-campus lectures are focused on problem-solving. Students will actively work on problems that cover the entire content of the course and the various engineering skills.</p> <p>Solutions will be discussed during the lectures.</p> <p>Homework for every lecture includes: watching the relevant video lectures dedicated to the upcoming course and the remaining exercises of the previous lecture.</p>                                                                                                                                                                                                                                              |   |
| <b>Literature and Study Materials</b>  | <p>Literature and Study Materials are provided on Brightspace:</p> <p>Videolectures<br/>Lectureslides.</p> <p>Large set of problems &amp; exercises including answers. Answers are posted on Brightspace after the problems have been discussed in the lectures.</p> <p>Large set of examples of exams and answers.</p> <p>The book 'Solar Energy, basics, technology and systems' is free to download as e-book via all online retailers. Hardcopies of the book are sold at the ETV counter (building 36).</p>                                                                                                                                                                                                                                                                                            |   |
| <b>Assessment</b>                      | <p>The course ET3034TU Solar Energy will be assessed in Q2 and in Q3 (resit) of the academic year 2022/2023 as follows:</p> <p>The exam is a written exam.<br/>The exam takes 3 hours.<br/>The exam consists of open questions and multiple choice questions.<br/>A formula sheet will be provided during the exam (example is posted on Brightspace).<br/>You can use only pen, paper and a calculator.<br/>The exam will be assessed with a mark with a resolution of 0.5.<br/>To pass the exam a score equal to or higher than 58.0 out of 100.0 is needed.</p>                                                                                                                                                                                                                                          |   |
| <b>Remarks</b>                         | Coordinator: Arno Smets                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |



| WB3575                                 | Energy Conversion: Devices, Systems and Efficiencies                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4 |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr.ir. T.M.J. Nijssen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>Responsible Instructor</b>          | M. Ramdin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Exam Period</b>                     | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Expected prior knowledge</b>        | Basic thermodynamics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |
| <b>Course Contents</b>                 | <p>This course will introduce students to the thermodynamics of energy conversion processes such as steam power plants, gas/wind turbines, electrochemical/fuel cells, and heat pumps. Thermodynamics will be applied to evaluate the efficiency and performance of energy conversion processes. In addition, devices and components used for energy conversion and storage will be discussed.</p> <p>The course includes the following lecturing elements:</p> <ul style="list-style-type: none"> <li>A review of basic thermodynamics, including 1st and 2nd law, energy, entropy, and exergy balances, and properties of mixtures</li> <li>Entropy loss of real processes</li> <li>Exergy loss of real processes</li> <li>Energy Conversion</li> <li>CO2 Capture and conversion</li> <li>Solar Energy, Hydrogen and Fuel Cells</li> <li>Economic evaluation of energy conversion processes</li> </ul> |   |
| <b>Study Goals</b>                     | <p><b>Study Goals</b></p> <p>At the end of the course, you should be able to describe the thermodynamics related to energy conversion. You will be able to apply entropy and exergy analysis to energy conversion systems, devices, and processes. Finally, you will be able to describe the operating principles of energy conversion processes and evaluate their economics.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
| <b>Education Method</b>                | (Online) lectures, video's, ungraded homework and (online) tutoring sessions, and graded group exercises                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
| <b>Books</b>                           | Moran & Shapiro: Principles of Engineering Thermodynamics, 9th Edition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
| <b>Assessment</b>                      | The final grade will be based on the group exercises (40%) and written exam (60%). The minimum grade for the written exam should be at least 5.5. Participation in the group exercises is mandatory to pass the course.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Department</b>                      | 3mE Department Process & Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |

| WB3580                           | Energy Storage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 5 |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| Responsible Instructor           | Prof.dr.ir. W. de Jong                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| Instructor                       | Dr. R. Delfos                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |
| Instructor                       | Dr.ir. J.M. Bloemendal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| Instructor                       | Prof.dr. P.J. Vardon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| Contact Hours / Week<br>x/x/x/x  | 4/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| Education Period                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| Start Education                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| Exam Period                      | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| Course Language                  | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| Course Contents                  | Energy storage becomes increasingly important in society due to the introduction of increased shares of fluctuating renewable energy sources, like wind and solar energy. Moreover, a decreased dependency on fossil fuel resources is aimed at.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |
|                                  | This course will introduce the students to the different strategies and systems of energy storage (mechanical, thermal, chemical), with an emphasis on large-scale storage systems for electricity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
|                                  | The course includes the following lecturing elements:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
|                                  | Introduction: context of renewable power supply (current and future) and scope.<br>Mechanical storage systems (pumped hydro-electricity, compressed air energy storage, flywheels).<br>Electrical storage systems (supercapacitors, superconducting magnetic Energy Storage).<br>Electrochemical storage systems (batteries, flow batteries)<br>Chemical storage systems (hydrogen production and storage options, power-to-gas, power-to-liquids, carbon sources).<br>Carbon capture and storage/utilization.<br>Thermal storage systems (sensible heat, phase change materials, thermo-chemical), including geothermal storage systems.                                                                                                        |   |
| Study Goals                      | The student can understand, select, and conceptually design and optimize processes for energy storage (both electricity and heat storage systems).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |
|                                  | More specifically, the student must be able to:<br>1. understand power supply and demand characteristics in relation to intermittent renewable energy sources and the role of energy storage herein;<br>2. understand and select appropriate systems for energy storage in terms of capacity and power and start-up and shut-down aspects;<br>3. solve problems related to mechanical, thermal, thermo-mechanical, electrical and chemical energy storage components of energy storage systems based on in depth knowledge of thermodynamics, mass and heat transfer, and (fluid) mechanics;<br>4. design basic process features for the abovementioned classes of energy storage systems.                                                       |   |
| Education Method                 | Plenary lectures including cases and solving problems (4 hours per week) and separate working classes (2 hours per week); all foreseen to be on-line.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| Computer Use                     | the students have to use standard computer tools to calculate thermodynamic properties and performance of simple reaction and separation system                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| Literature and Study Materials   | Optional books:<br>O.S. Burheim (2017) Engineering Energy Storage, Academic Press, 1st ed.<br>G.Genta (1985), Kinetic Energy Storage, Butterworth<br>R.A. Huggins (2010), Energy Storage, Springer Verlag<br>A.F. Mills, (Basic) Heat (and Mass) Transfer, Pearson (all editions) or alike<br>M.J. Moran, H.N. Shapiro, D.D. Boettner, and M.B. Bailey Principles of engineering thermodynamics (all editions)<br>P.K. Nag (2014), Power Plant Engineering, McGraw-Hill, 4th ed.<br>A.G. Ter-Gazarian (2010), Energy Storage for Power Systems, 2nd. ed.<br>Towler and Sinnott, Chemical Engineering Design Principles, Practice and Economics, second edition.<br>A. Vieira da Rosa (2013), Fundamentals of renewable energy processes, 3rd ed. |   |
|                                  | Articles:<br>Ausfelder et al. (2015) Energiespeicherung als Element einer sicheren Energieversorgung, Chemie Ingenieur Technik, 87(1-2), pp. 17-89.<br>Lopes Ferreira et al. (2013) Characterisation of electrical energy storage technologies, Energy, 53, 288-298.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| Assessment                       | Written exam (on-line; individual), 75% of the final grade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
|                                  | Assignment (group) on thermal energy storage systems, 25% final grade                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| Permitted Materials during Tests | A standard (non programmable) calculator                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |
| Percentage of Design             | 35%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |
| Department                       | 3mE Department Process & Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |

| WB3585                                  | Assessment of Energy Systems                                                                                                                                                                                                                                                                                                                    | 3 |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>           | Dr.ir. L. Stougie                                                                                                                                                                                                                                                                                                                               |   |
| <b>Instructor</b>                       | Dr. L.L. Cutz IJchajchal                                                                                                                                                                                                                                                                                                                        |   |
| <b>Instructor</b>                       | Dr.ir. G. Korevaar                                                                                                                                                                                                                                                                                                                              |   |
| <b>Contact Hours / Week</b><br>x/x/x/x  | 0/4/0/0                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Education Period</b>                 | 2                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Start Education</b>                  | 2                                                                                                                                                                                                                                                                                                                                               |   |
| <b>Exam Period</b>                      | 2<br>3                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Course Language</b>                  | English                                                                                                                                                                                                                                                                                                                                         |   |
| <b>Course Contents</b>                  | This course introduces systematic ways to evaluate energy conversion and storage systems; life cycle assessment (LCA) and exergy analysis are applied to cases.                                                                                                                                                                                 |   |
| <b>Study Goals</b>                      | 1) use thermodynamic knowledge, and especially the concept of exergy, in the field of large scale energy conversion and storage systems<br>2) perform a system assessment with the help of energy, exergy and LCA concepts and tools<br>3) evaluate large scale energy conversion and storage systems and make recommendations to improve these |   |
| <b>Education Method</b>                 | Lectures and case studies                                                                                                                                                                                                                                                                                                                       |   |
| <b>Literature and Study Materials</b>   | Reader                                                                                                                                                                                                                                                                                                                                          |   |
| <b>Assessment</b>                       | The exam is on campus in the form of a closed book written exam. In case on campus education and examination is not possible, the exam will be in the form of a remote written exam with several fraud prevention measures.                                                                                                                     |   |
| <b>Permitted Materials during Tests</b> | Simple, non-programmable calculator                                                                                                                                                                                                                                                                                                             |   |
| <b>Tags</b>                             | Energy<br>Energy & Industry<br>Thermodynamics                                                                                                                                                                                                                                                                                                   |   |

| WB3595                                 | Design Project Renewables Based Energy Conversion and Storage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 12 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <b>Responsible Instructor</b>          | Dr.ir. J.T. Padding                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |
| <b>Contact Hours / Week</b><br>x/x/x/x | x/x/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |
| <b>Education Period</b>                | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |
| <b>Exam Period</b>                     | Different, to be announced                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |    |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |
| <b>Course Contents</b>                 | <p>The finalisation of the ELECS minor consists of a project (12 ECTS per student) in which the students, in interdisciplinary group work, based on the boundary conditions of a renewable energy supply system, will conceptually develop, design and evaluate an energy conversion system with storage, or will develop and test a novel energy conversion/storage concept on a component level. This can be done via modelling and/or setting up and performing experiments.</p> <p>The design project will take place during the 1st and 2nd period (Q1 and Q2) of the academic year. During Q1 (3 ECTS) the students will more precisely define the goal of the design, do a literature search, and do preparatory work (including ordering materials, parts and sensors in case of experimental setups). Q1 will be closed with an oral presentation of the design plan. The actual design and evaluation of the process or component will take place during Q2 (9 ECTS). Q2 will be closed with a final report, a working model or prototype, and a pitch and poster presentation at the ELECS final symposium.</p>                                                                                                                                                                                                                                                                                                                |    |
| <b>Study Goals</b>                     | <p>After completion of the course, the students are able to:</p> <ul style="list-style-type: none"> <li>- apply and extend the fundamental knowledge that plays an important role in sustainable energy conversion and storage (in particular of thermodynamics, heat transfer, reaction kinetics and fluid mechanics) in designing a process.</li> <li>- divide a given process in sub-processes, together forming the entire process</li> <li>- apply thermodynamics to derive a process design of a technical system and to optimize the thermodynamic design (not for all projects).</li> <li>- apply heat transfer concepts to determine the main dimensions of heat exchangers (not for all projects)</li> <li>- apply reaction kinetics concepts to derive the main dimensions of reactors (not for all projects)</li> <li>- develop a list of requirements for the process design, accounting for its sustainability and limits to ethical, societal and safety</li> <li>- evaluate a design according to sustainability criteria which have to be formulated</li> <li>- write a well-structured technical report</li> <li>- in case of an experimental design project: deliver a working prototype</li> <li>- in case of a simulation design project: deliver a working model in Matlab, Python, or a programming language like C or Java</li> <li>- explain and defend their design in an oral (poster) presentation</li> </ul> |    |
| <b>Education Method</b>                | During 14 weeks, students work in small groups of 4 to 5 students on a design project. The projects will be supervised by permanent staff of the research groups contributing to the minor. Dedicated instructions will be given on aspects of modelling and simulation and/or laboratory practices (a.o. safety).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |
| <b>Assessment</b>                      | <p>In Q1, week 4, the students need to deliver a 2-to-3-page summary of the literature search. This will be evaluated by the supervisors (weight 10%).</p> <p>In Q1, week 9, the students need to deliver a 8-minute design plan presentation (i.e. the motivation, precise goal, results of the literature search, and an outline of their approach), followed by 3 minutes of questions. The presentations of all student groups will be organised in a joint session with as many supervisors present as possible. The presentation will be evaluated by the supervisors (weight 15%).</p> <p>In Q2, week 9, the students need to deliver a final report and a working model or prototype. This will be evaluated by the supervisors (report has weight 40%, model or prototype has weight 20%).</p> <p>A few days later in Q2, week 9, during the ELECS final symposium, the students will give a short pitch of their design in 4 minutes, followed by presentation of more details in a poster session. The pitch and poster presentation will be evaluated by the supervisors (weight 15%).</p>                                                                                                                                                                                                                                                                                                                                    |    |
| <b>Department</b>                      | 3mE Department Process & Energy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |    |

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|---------------------|-------------------------------------------------------|
| <b>Year</b>         | <b>2023/2024</b>                                      |
| <b>Organization</b> | <b>Mechanical, Maritime and Materials Engineering</b> |
| <b>Education</b>    | <b>Minors Mechanical Engineering</b>                  |

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|------------------------------------------------------|
| <b>WB-MI-238-23 Minor Authenticity and Art Crime</b> |
|------------------------------------------------------|

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|----------------------------------------|------------------------------------------------|----------|
| <b>MS3310UL</b>                        | <b>What is Authenticity in Art</b>             | <b>5</b> |
| <b>Responsible Instructor</b>          | M. Mira                                        |          |
| <b>Contact Hours / Week</b><br>x/x/x/x | 2/0/0/0                                        |          |
| <b>Education Period</b>                | 1                                              |          |
| <b>Start Education</b>                 | 1                                              |          |
| <b>Exam Period</b>                     | 1<br>2                                         |          |
| <b>Course Language</b>                 | English                                        |          |
| <b>Department</b>                      | 3mE Department Materials Science & Engineering |          |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |
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| <b>MS3320</b>                          | <b>The Biography of Art Objects: Understanding the Material Life Cycle of Artwork through Physical and</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>5</b> |
| <b>Responsible Instructor</b>          | Dr. M.W.E.M. Alfeld                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |
| <b>Instructor</b>                      | Dr. L. van Eijck                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |
| <b>Instructor</b>                      | Prof.dr. J. Dik                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |
| <b>Instructor</b>                      | Dr.ir. D.J.M. Ngan-Tillard                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |
| <b>Instructor</b>                      | Dr.ir. D.J.M. Ngan-Tillard                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |
| <b>Contact Hours / Week</b><br>x/x/x/x | 2/0/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |
| <b>Education Period</b>                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |
| <b>Start Education</b>                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |
| <b>Exam Period</b>                     | 1<br>2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |
| <b>Course Contents</b>                 | <p>An artwork is often valued primarily on account of its attribution to a specific creator or location of origin, date or period, and provenance, i.e. not only on the basis of its physical appearance. While an attribution may provide an historical context to the work, establish quality and uniqueness, all these subjectively meaningful parameters are undermined should the work turn out to be a forgery.</p> <p>When authenticating art, experts eventually decide what is of cultural significance and what is not. Authenticity has concrete repercussions on the market, in the form of its influence on an artworks economic value, and in law, given the liability of experts and sellers for misattributions and the sale of forgeries. For the past two decades the art world has increasingly relied on forensics, i.e. the dedicated chemical and physical investigation of artwork to help answer questions of authenticity. Nowadays, the public museum field, but also the commercial markets such as auction houses and international art fairs involve forensic analysis in their vetting procedure. The expert opinion of the art expert (connoisseurship), the line of ownership of an artwork (the provenance), and materials analysis (forensics) are currently the three most important pillars to establish the authenticity of an artwork.</p> <p>The materials of artworks offer a unique insight into the making of art, covering the entire lifecycle of an object. Raw materials, their geographic and chronological origin, artist's working techniques, material decay, restoration treatments and re-use and recycling of art objects are all part of the artwork lifecycle. Understanding these aspects is key to understanding the artwork's biography, its provenance and, ultimately, the artwork's authenticity.</p> <p>The fundamentals of this life cycle will be presented in an introductory lecture and later illustrated throughout this course in case studies. The main focus of the introduction will be on paint and painted objects and the methods used for the investigation of historical paintings (UV, IRR, XRR).</p> <p>The tremendous methodological progress in non-destructive analysis and chemical and physical imaging diagnostics achieved in the past decades will be illustrated in a series of lectures, which take the variable entry level of the students into account. The lectures will start from the nature of matter (atoms, their cores and electron shells) and radiation, especially X-rays. This basic instruction will be consolidated by peer instruction in the project groups.</p> <p>The production and detection of X-rays and their interaction with matter will be presented and it will be explained how these are used to acquire computed tomography data. The value this non-invasive 3D technique has to gain insight into an objects lifecycle will be illustrated on pottery/ clay cuneiform tablets, stone (beads/ flints), glass (flasks and beads) and corroded metal (coins/ filigree sphere) objects.</p> <p>X-ray spectroscopy and diffraction will be discussed at a basic level and the use of these basic principles in MA-XRF and MA-XRD imaging will be illustrated in studies of pigments of 17th century paintings and archaeological objects from classical antiquity (Etruscan and Greek sites).</p> <p>The individual techniques will be concluded with an introduction to neutrons and their use in neutron radiography, tomography, gamma spectroscopy and neutron activation analysis with application examples.</p> <p>This series of lectures will be closed by two events: the students will visit the TU Delft campus and travel in groups to see instruments that are used for the techniques described above. Finally, in a formative test the student groups need to identify the techniques most suitable to address a series of given problems.</p> |          |
| <b>Study Goals</b>                     | <p>After this course the student is capable to:</p> <p>Analyze the challenges presented by potential forgeries by interpreting different knowledge sources.</p> <p>Discuss the physical and chemical principles behind common analysis techniques and the build-up of artworks.</p> <p>Formulate the materials science questions for an art object.</p> <p>Propose suitable methods for investigating these questions.</p> <p>Interpret the acquired data to contribute to an answer in the discussion of a potential forgery.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |
| <b>Education Method</b>                | <p>Lectures</p> <p>Site and Laboratory visits</p> <p>Supervised group project on a case study</p> <p>The students will document all these events in a portfolio.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |
| <b>Assessment</b>                      | The grades of the students will be derived from their portfolio.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |
| <b>Department</b>                      | 3mE Department Materials Science & Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |

|                                        |                                                |          |
|----------------------------------------|------------------------------------------------|----------|
| <b>MS3330EUR</b>                       | <b>Crime and Disruption in the Art Market</b>  | <b>5</b> |
| <b>Responsible Instructor</b>          | M. Mira                                        |          |
| <b>Contact Hours / Week</b><br>x/x/x/x | 2/0/0/0                                        |          |
| <b>Education Period</b>                | 1                                              |          |
| <b>Start Education</b>                 | 1                                              |          |
| <b>Exam Period</b>                     | 1<br>2                                         |          |
| <b>Course Language</b>                 | English                                        |          |
| <b>Department</b>                      | 3mE Department Materials Science & Engineering |          |

|                                        |                                                                           |           |
|----------------------------------------|---------------------------------------------------------------------------|-----------|
| <b>MS3340UL</b>                        | <b>Authenticity and Art Crime: A Challenging Case Study (Leiden part)</b> | <b>10</b> |
| <b>Responsible Instructor</b>          | M. Mira                                                                   |           |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/4/0/0                                                                   |           |
| <b>Education Period</b>                | 2                                                                         |           |
| <b>Start Education</b>                 | 2                                                                         |           |
| <b>Exam Period</b>                     | 2<br>3                                                                    |           |
| <b>Course Language</b>                 | English                                                                   |           |
| <b>Department</b>                      | 3mE Department Materials Science & Engineering                            |           |

| MS3350                                 | Authenticity and Art Crime: A Challenging Case Study (Delft part)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 5 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| <b>Responsible Instructor</b>          | Dr. M.W.E.M. Alfeld                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |
| <b>Contact Hours / Week</b><br>x/x/x/x | 0/2/0/0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Education Period</b>                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Start Education</b>                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Exam Period</b>                     | 2<br>3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
| <b>Course Language</b>                 | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Course Contents</b>                 | <p>An artwork is often valued primarily on account of its attribution to a specific creator or location of origin, date or period, and provenance, i.e. not only on the basis of its physical appearance. While an attribution may provide an historical context to the work, establish quality and uniqueness, all these subjectively meaningful parameters are undermined should the work turn out to be a forgery.</p> <p>When authenticating art, experts eventually decide what is of cultural significance and what is not. Authenticity has concrete repercussions on the market, in the form of its influence on an artworks economic value, and in law, given the liability of experts and sellers for misattributions and the sale of forgeries. For the past two decades the art world has increasingly relied on forensics, i.e. the dedicated chemical and physical investigation of artwork to help answer questions of authenticity. Nowadays, the public museum field, but also the commercial markets such as auction houses and international art fairs involve forensic analysis in their vetting procedure. The expert opinion of the art expert (connoisseurship), the line of ownership of an artwork (the provenance), and materials analysis (forensics) are currently the three most important pillars to establish the authenticity of an artwork.</p> <p>This class will build on the previous class (MS3320) and consolidate its content by Challenge Based Learning. This means that based on a real-life thesis, students and teachers learn how to deal with multi-faceted practical challenges and engage with them in experimental and practical ways across disciplines and universities and together with societal stakeholders in the cultural heritage field. Research questions will be answered through the phases of Challenge Based Learning: Engage, Investigate and Act. Students will be guided by content and field experts and by methodological experts in the different phases of the Challenge Based Learning process. This will not only teach them how to put their recently obtained knowledge to use, but it may also forge a way of seeing where there is need of improvement in terms of their approach. The course will consist of weekly seminars (each three hours) and a number of fieldtrips. Through a series of assignments, the students will deepen their knowledge and sharpen their skills by collaborating in small groups. In phase two (CBL Investigate) each week, the students will receive small assignments which will help them analyze and discuss the case-study under investigation. These small assignments will focus on a specific aspect or discipline to be considered while doing these kinds of investigations (e.g. one session and assignment will focus on a particular type of visual or chemical analysis or provenance research).</p> <p>Lastly, in phase three (CBL Act) the students will be encouraged to actively use and apply their developed method to the case-study. In the end, they will formulate possible solutions, conclusions and future research based upon their experiences. This will be presented in the final meeting during a 20 minute talk per group. They will also hand in their Portfolio including the different and a reflection on their collaborations across different disciplines.</p> |   |
| <b>Study Goals</b>                     | <p>After this course the student is capable to:</p> <ul style="list-style-type: none"> <li>Formulate relevant research questions and select relevant research techniques.</li> <li>Perform in-depth visual analysis with the naked eye with a focus on the case study investigated in class.</li> <li>Perform in-depth analysis with relevant technical research methods with a focus on the case study investigated in class.</li> <li>Undertake provenance research using various data bases.</li> <li>Recognize and search for relevant patterns with a view to spotting potential counterfeits.</li> <li>Collaborate across disciplines.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |
| <b>Education Method</b>                | <ul style="list-style-type: none"> <li>Plenary sessions</li> <li>Excursions to master classes and laboratories</li> <li>Weekly assignments and group work on a project in a group</li> <li>The students will document all these events in a portfolio.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| <b>Assessment</b>                      | The grades of the students will be derived from their portfolio (70%) and the group presentations (30%).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| <b>Tags</b>                            | <ul style="list-style-type: none"> <li>Chemistry</li> <li>Diverse</li> <li>Ethics</li> <li>Group work</li> <li>Image processing</li> <li>Physics</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
| <b>Department</b>                      | 3mE Department Materials Science & Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |



## Dr. M.W.E.M. Alfeld

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Team Matthias Alfeld           |
| <b>Telephone</b>  | +31 15 27 88253                |

## Dr.ir. A. van Beek

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Mechatronic Systems Design     |
| <b>Telephone</b>  | +31 15 27 86984                |

## Dr.ir. J.M. Bloemendal

|                   |                 |
|-------------------|-----------------|
| <b>Department</b> | Water Resources |
| <b>Telephone</b>  | +31 15 27 89916 |
| <b>Room</b>       | 23.HG 4.82      |

## Ir. C.C. Claij

|                   |                            |
|-------------------|----------------------------|
| <b>Department</b> | Support Cognitive Robotics |
| <b>Telephone</b>  | +31 15 27 87603            |
| <b>Room</b>       | 34.F-2-360                 |

## Dr. L.L. Cutz IJchajchal

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Large Scale Energy Storage     |
| <b>Telephone</b>  | +31 15 27 85797                |
| <b>Room</b>       | 34b.K-1-140                    |

## Dr. R. Delfos

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Energy Technology              |
| <b>Telephone</b>  | +31 15 27 82963                |
| <b>Room</b>       | 34b.K-0-270                    |

## Prof.dr. J. Dik

|                   |                                   |
|-------------------|-----------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng    |
| <b>Department</b> | Materials Science and Engineering |
| <b>Telephone</b>  | +31 15 27 89571                   |

## T. Durieux

|                   |                      |
|-------------------|----------------------|
| <b>Department</b> | Software Engineering |
| <b>Room</b>       | 28.1.W780            |

## Dr. L. van Eijck

|                   |                                               |
|-------------------|-----------------------------------------------|
| <b>Unit</b>       | Technische Natuurwetenschappen                |
| <b>Department</b> | RST/Neutron and Positron Methods in Materials |
| <b>Telephone</b>  | +31 15 27 85731                               |
| <b>Room</b>       | 50.02.01.190                                  |

## Prof.dr.ir. W. de Jong

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Large Scale Energy Storage     |
| <b>Telephone</b>  | +31 15 27 89476                |

## M. Klomp

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Robot Dynamics                 |
| <b>Telephone</b>  | +31 15 27 86740                |
| <b>Room</b>       | 34.F-2-420                     |

## Dr.ir. G. Korevaar

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Techniek, Bestuur & Management |
| <b>Department</b> | Energie and Industrie          |
| <b>Telephone</b>  | +31 15 27 83659                |

|             |           |
|-------------|-----------|
| <b>Room</b> | 31.A3.200 |
|-------------|-----------|

## I.D. van Kreveld

|                   |                                              |
|-------------------|----------------------------------------------|
| <b>Department</b> | Computer Science & Engineering-Teaching Team |
| <b>Room</b>       | 28.2.E400                                    |

## Dr.ir. N.P. van der Meijs

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform. |
| <b>Department</b> | Signal Processing Systems      |
| <b>Telephone</b>  | +31 15 27 86258                |
| <b>Room</b>       | 36.HB 17.310                   |

## M. Mira

|                   |                |
|-------------------|----------------|
| <b>Department</b> | Team Joris Dik |
|-------------------|----------------|

## F. Mulder

|                   |                                              |
|-------------------|----------------------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform.               |
| <b>Department</b> | Computer Science & Engineering-Teaching Team |
| <b>Telephone</b>  | +31 15 27 83184                              |
| <b>Room</b>       | 28.2.E340                                    |

## Dr.ir. D.J.M. Ngan-Tillard

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Geo-engineering                |
| <b>Telephone</b>  | +31 15 27 83325                |
| <b>Room</b>       | 23.KG 00.140                   |

## Ir. A.M. van der Niet

|                   |                            |
|-------------------|----------------------------|
| <b>Department</b> | Support Cognitive Robotics |
| <b>Room</b>       | 34.F-2-420                 |

## Dr.ir. T.M.J. Nijssen

|                   |                            |
|-------------------|----------------------------|
| <b>Department</b> | Engineering Thermodynamics |
| <b>Room</b>       | 34b.K-1-260                |

## Dr.ir. R.A.J. van Ostayen

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Mechatronic Systems Design     |
| <b>Telephone</b>  | +31 15 27 81647                |
| <b>Room</b>       | 34.G-1-460                     |

## Dr.ir. J.T. Padding

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Complexe Vloeistof Processen   |
| <b>Telephone</b>  | +31 15 27 86605                |
| <b>Room</b>       | 34b.K-1-200                    |

## G. Papaioannou

|                   |                      |
|-------------------|----------------------|
| <b>Department</b> | Intelligent Vehicles |
| <b>Telephone</b>  | +31 15 27 89329      |
| <b>Room</b>       | 34.E-0-220           |
| <b>Department</b> | Intelligent Vehicles |
| <b>Telephone</b>  | +31 15 27 89329      |
| <b>Room</b>       | 34.E-0-220           |

## M. Ramdin

|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Mech, Maritime & Materials Eng |
| <b>Department</b> | Engineering Thermodynamics     |
| <b>Room</b>       | 34b.K-1-290                    |

Dr.ing. M.C. Rozendaal

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|                   |                                        |
|-------------------|----------------------------------------|
| <b>Unit</b>       | Industrieel Ontwerpen                  |
| <b>Department</b> | Human Information Communication Design |
| <b>Telephone</b>  | +31 15 27 86947                        |
| <b>Room</b>       | 32.C-3-250                             |

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Prof.dr.ir. A.H.M. Smets

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|                   |                                    |
|-------------------|------------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform.     |
| <b>Department</b> | Photovoltaic Materials and Devices |
| <b>Telephone</b>  | +31 15 27 88739                    |
| <b>Room</b>       | 36.LB 03.420                       |

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Dr.ir. L. Stougie

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|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Techniek, Bestuur & Management |
| <b>Department</b> | Energie and Industrie          |
| <b>Telephone</b>  | +31 15 27 87041                |
| <b>Room</b>       | 31.a3.320                      |

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Dr. D.A. von Terzi

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|                   |                 |
|-------------------|-----------------|
| <b>Department</b> | Wind Energy     |
| <b>Telephone</b>  | +31 15 27 83379 |

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Prof.dr. P.J. Vardon

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|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Civiele Techniek & Geowetensch |
| <b>Department</b> | Geo-engineering                |
| <b>Telephone</b>  | +31 15 27 81456                |
| <b>Room</b>       | 23.KG 00.490                   |

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Dr.ir. A.C. Viré

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|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Luchtvaart- & Ruimtevaarttechn |
| <b>Department</b> | Wind Energy                    |
| <b>Telephone</b>  | +31 15 27 81385                |
| <b>Room</b>       | 62.5.19                        |

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Prof.dr. A.E. Zaidman

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|                   |                                |
|-------------------|--------------------------------|
| <b>Unit</b>       | Elektrotechn., Wisk. & Inform. |
| <b>Department</b> | Software Engineering           |
| <b>Telephone</b>  | +31 15 27 84385                |
| <b>Room</b>       | 28.1.W860                      |

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